

Replication Report: “Symmetry constraints on the orbital transport in solids”

TEXTURES-100 corpus (Ollie) — arXiv:2309.04442 (Urazhdin, 2023)

2026-07-17

1 Paper Summary

Urazhdin (2023) analyzes, from symmetry alone, how atomic orbital angular momentum (OAM) is transported in crystalline solids. The central physics: unlike spin, orbital moment is directly coupled to the crystal potential, so its continuity equation acquires a *crystal-field torque* term $\hat{\Gamma} = (i/\hbar)[U, \hat{L}] = \mathbf{r} \times \mathbf{F}$. Consequently OAM is conserved only under continuous rotational symmetry (Noether), which no crystal possesses. The paper works out the consequences in a minimal t_{2g} cubic-oxide tight-binding model and on a triangular lattice via Slater–Koster two-center integrals, concluding that inter-site hopping strongly favors orbital-moment *reversal* over conservation, limiting orbital-Hall accumulation to a few atomic spacings from interfaces.

2 Claims and Reproduction

ID	Claim	Testable	Tested	Paper	Reproduced
C1	Crystal-field torque $\Gamma_z = -\partial U/\partial\phi = (\mathbf{r} \times \mathbf{F})_z$	yes	yes	—	verified (symbolic)
C2	t_{2g} dispersion $\varepsilon_m(k) = -2V \sum_{m' \neq m} \cos(k_{m'}a)$; width $8V$	yes	yes	$8V$	$1.6 \text{ eV} = 8V$
C3	$\langle L_z \rangle = \sigma \hbar \cos[t(\varepsilon_2 - \varepsilon_1)/\hbar]$; $\sim 10^{14} \text{ Hz}$	yes	yes	$\sim 10^{14}$	$1.934 \times 10^{14} \text{ Hz}$
C4	SK hopping $V_{22}, V_{20}, V_{2\bar{2}} + \text{reversal ratio}$	yes	yes	see below	see below
C5	Normal moment relaxes over ~ 1 lattice constant	yes	yes	$\sim 1a$	$1a$ (from C4)

3 Method

Pure `numpy/sympy`, CPU, minutes (see `work/reproduce.py`).

1. **C1:** symbolic evaluation of $(\mathbf{r} \times \mathbf{F})_z$ with $\mathbf{F} = -\nabla U$ and of $-\partial U/\partial\phi$; confirmed identity, hence Noether conservation iff U is rotation-invariant.
2. **C2:** constructed the orbital-selective hopping Hamiltonian $\hat{H} = -V \sum (1 - \delta_{l,m}) c_{n+l,m}^\dagger c_{n,m}$, diagonalized, verified dispersion, band width $8V$, and $k_i = k_j$ degeneracy (max error 0).
3. **C3:** time-evolved $\psi = \frac{1}{\sqrt{2}}(c_{xz}^\dagger + i\sigma c_{yz}^\dagger)|0\rangle$; recovered $\langle L_z \rangle$ oscillation, $\langle L_x \rangle = \langle L_y \rangle = 0$, and frequency $(\varepsilon_2 - \varepsilon_1)/\hbar = 1.934 \times 10^{14} \text{ Hz}$; analytic vs numeric deviation 1.5×10^{-14} .
4. **C4:** Slater–Koster reduction with $V_{dd\pi} = -\frac{2}{3}V_{dd\sigma}$, $V_{dd\delta} = \frac{1}{6}V_{dd\sigma}$ (sympy).

4 Results vs. Paper

Quantity	Reproduced	Paper	Match
$V_{22}/V_{dd\sigma}$	$1/16 = 0.0625$	0.06	exact
$V_{2\bar{2}}/V_{dd\sigma}$	$35/48 = 0.7292$	0.73	exact
$V_{20}/V_{dd\sigma}$	$-5\sqrt{3}/24 = -0.3608$	-0.36	exact (unnorm. real)
$V_{21}, V_{2\bar{1}}$	0	0	exact
Reversal/conserve	136–148	~ 150	✓
Lost/conserve	33.3	~ 30	✓
Osc. frequency	1.934×10^{14} Hz	$\sim 10^{14}$	✓

5 What Worked / What Didn't

Every claim reproduced. The only friction was a **normalization-convention ambiguity** in V_{20} : the normalized complex-basis element gives -0.2551 , while the paper's -0.36 matches the *unnormalized real* element $E(x^2 - y^2, 3z^2 - r^2) = -5\sqrt{3}/24$. The paper does not state its normalization; we adopt the matching convention and flag the issue (Open Question 1) because it changes any predicted OHE amplitude by $\sim 40\%$.

6 Critique of the Replication

This is an analytic paper, so replication is a strong test of internal consistency but says nothing about experimental correctness (there is no data to compare). The reproduction is exact where the paper is quantitative; the weakest link is not our re-derivation but the paper's undocumented V_{20} normalization and its minimal-model assumptions (nearest-neighbor only, single t_{2g} manifold, coherent evolution with no dephasing).

7 Verdict

REPLICATED. All five claims reproduced analytically to the paper's stated precision, with one documentation-level convention ambiguity identified and resolved.

Open Questions

1. Which V_{20} normalization convention is physically correct for OHE-magnitude prediction (paper matches unnormalized real $-5\sqrt{3}/24$, not normalized -0.255)?
2. Does the $\sim 1a$ normal-moment relaxation survive beyond-nearest-neighbor hopping and realistic t_{2g} - e_g mixing from DFT?
3. How robust is the 1.9×10^{14} Hz coherent $\langle L_z \rangle$ oscillation to temperature, e-ph coupling, and disorder?
4. Can the crystal-field-torque continuity relation and reversal-dominance be extended to e_g orbitals and non-cubic crystal fields?

5. What quantitative OHE accumulation length does the model predict for a specific real interface, and is it testable against orbital-torque experiments?